Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

David Afonso Valente *dvalente@bottlecapai.com*
Bottlecap AI

Abstract

Base models are chosen for fine-tuning by their benchmark scores. We test a label-free alternative: score each candidate's bits per byte on whole documents of fresh, unlabelled text from the target domain. For fifteen base models fine-tuned on a news-writing task, this ranking places every model within one rank of where it finishes and orders 37 of 39 size-matched pairs as the fine-tuned systems do. GSM8K and MMLU-Pro order the same pairs at chance, and so do the models' own scores on the task before any training: fine-tuning reorders the models, and the document reading predicts the order they end in. HellaSwag orders 34. Two details decide the result. The text must come from the task's domain, or close to it: the same measurement on arXiv mathematics orders 23 of the 39 pairs. And each document must be read whole, from its beginning: scored on fixed-length blocks cut from concatenated text, as perplexity is often computed, the ranking puts the model that fine-tunes best thirteenth of fifteen. Read this way, the measurement needs no labels, no task and no training, and adapting the models to the domain first barely changes it. Which benchmark the ranking resembles follows the text: HellaSwag on general prose, GSM8K and MMLU-Pro on mathematics. Code and artifacts: https://github.com/DavidAfonsoValente/entropy-bench.

1 Introduction

An engineer choosing a base model for a product usually has plenty of text from the product's domain and a shortlist of open checkpoints, and the model they pick will be fine-tuned before it ships. Fine-tuning every candidate would settle the choice, but it needs a task definition, labelled data and a training run per model. So the choice is made from benchmark tables. Model reports compare base models on MMLU-Pro (Wang et al., 2024), GSM8K (Cobbe et al., 2021), HellaSwag (Zellers et al., 2019) and similar suites (Yang et al., 2024), and the Open LLM Leaderboard ranks pretrained models on six benchmarks (Fourrier et al., 2024).

We find that the fine-tuned ranking can be read off before any task exists, from unlabelled text alone. For each candidate we compute its loss in bits per byte (BPB) on whole documents of fresh text from the domain, each document read from its own beginning. On a news-writing task, this ranking places all fifteen fine-tuned models within one rank of where they finish (Figure 1). GSM8K and MMLU-Pro misjudge several of the same models, and order size-matched pairs of them at chance. So do the models' own scores on the task before any training: fine-tuning reorders the models, and the document reading predicts the order they end in.

That held-out loss predicts fine-tuning is not the received view. The most direct test we know of fine-tunes fifty 1B models and finds that perplexity on a general held-out set orders pairs of the fine-tuned systems worse than chance, while few-shot accuracy on the task does best (Zeng et al., 2025). Besides the cohort, two details separate that measurement from ours, and each is often got wrong.

The first is which text. When we score the same fifteen models on arXiv mathematics instead of news, the ranking orders only 23 of 39 size-matched pairs as the news fine-tune does, against 37 with news text, and falls below ranking by model size (Section 5.4). Which benchmark the ranking agrees with follows the text too. On news, Reddit and Hacker News it agrees with HellaSwag; on mathematics it agrees more with GSM8K and MMLU-Pro.

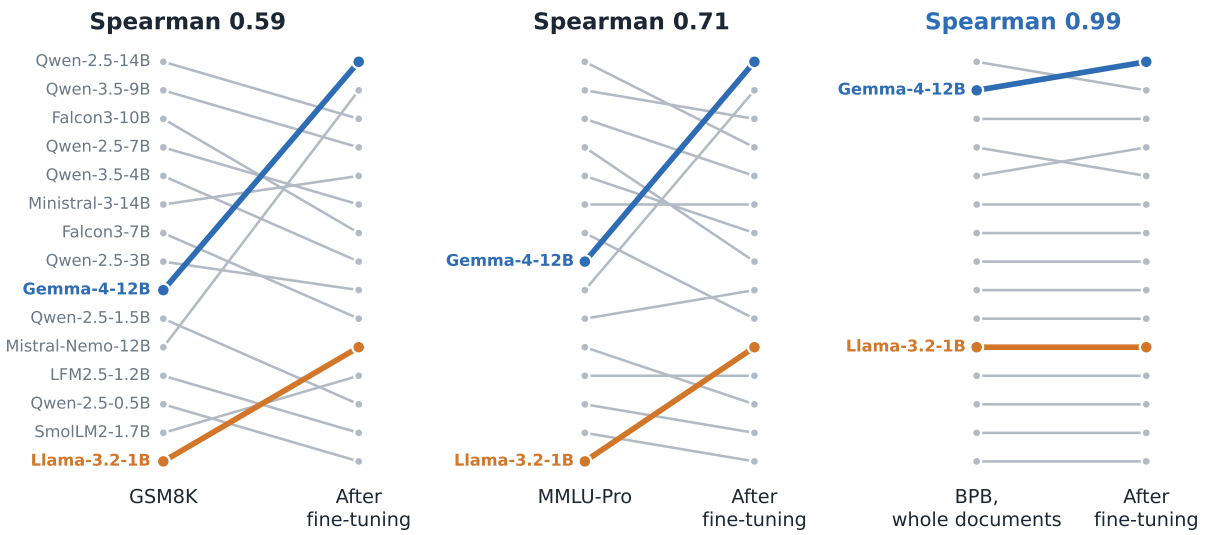


Figure 1: Which ranking, taken before fine-tuning, says how the models finish? Fifteen base models, each fine-tuned on the same news-writing task. In each panel a line joins a model's rank on a measure taken before fine-tuning (left) to its rank after fine-tuning, by ROUGE-L on 500 held-out articles (right); rank 1 is best. Crossing lines are models the measure misjudges. BPB on whole news documents, with no training, places every model within one rank of its finish, and eleven exactly. Gemma-4-12B (blue), ninth on GSM8K and eighth on MMLU-Pro, builds the best fine-tuned system; Llama-3.2-1B (orange), last on both, finishes eleventh.

The second is how the text is cut. A common convention joins the test set into one stream and scores fixed-length blocks cut from it (Hugging Face, 2024), so most blocks begin mid-document. Scored that way, the same news text misranks several strong models, and Gemma-4-12B, which builds the best fine-tuned system of the fifteen, comes thirteenth (Section 5.2). Reading each document from its own start fixes this with no training. Adapting every model briefly to the domain repairs the block reading as well, which is how we found the problem, but on whole documents it changes almost nothing (Section 5.3).

Loss is a natural measure for this choice. A base model acquires its capabilities by minimising cross-entropy on text, and fine-tuning minimises it again on the task. Cross-entropy is also a strictly proper scoring rule (Gneiting & Raftery, 2007). If p is the distribution that generated the domain's text and q is a model, the model's expected loss is

$$H(p, q) = H(p) + \mathrm{KL}(p \,\|\, q), \tag{1}$$

where $H(p)$, the entropy of the text, is the same for every model (Cover & Thomas, 2006). The difference between two models' held-out losses therefore estimates the difference in how far each is from the domain's text. Whether that predicts usefulness on a task is an empirical question, which Section 5.1 tests. We count the loss per byte, because each model has its own tokenizer but they all read the same bytes, and we compute it on fresh text: Reddit, Hacker News and arXiv were collected after every candidate was released, and the news snapshot, which predates some releases, is screened for membership (Section 3). Benchmarks offer neither guarantee: vendors optimise for the ones they report, and their questions can leak into pre-training data.

We evaluate seventeen base models from 0.5B to 35B parameters. Fifteen are fine-tuned on the news task, writing an article's opening paragraph from its body, and eleven are scored on all four corpora under the block readings too (Table 9). Our main findings are:

- **BPB on whole documents of the domain's text anticipates the fine-tuned ranking.** It orders 37 of 39 size-matched pairs as the fine-tuned systems do and places every model within one rank of its finish. HellaSwag orders 34; GSM8K, MMLU-Pro and the models' own untrained scores on the task are at chance (Section 5.1).

- **Blocks cut from a stream misrank strong models.** Scored on fixed-length blocks, Gemma-4-12B falls from second to thirteenth, and the ranking orders 30 of the 39 pairs (Section 5.2).
- **No training is needed.** Adapting every model to the domain before reading the documents lowers each model's loss but barely changes the order, on all four corpora (Section 5.3).
- **The text has to be like the task's.** Reddit text orders 34 of the 39 news pairs, Hacker News 31 and mathematics 23. Across corpora, the ranking agrees with HellaSwag on general text and with GSM8K and MMLU-Pro on mathematics (Section 5.4).

The measurement is deterministic: re-running it reproduces every model's BPB exactly (Section 5.5). All numbers in the paper can be recomputed from the released artifacts with `make check`:

`https://github.com/DavidAfonsoValente/entropy-bench`

The paper and its artifacts are also attached to each release (`https://github.com/DavidAfonsoValente/entropy-bench/releases/latest`).

2 Related Work

How base models are chosen. Base models are released and compared through benchmark tables. The Qwen-2.5 report, for one, compares its base models with others on MMLU, MMLU-Pro, GSM8K, HellaSwag and a dozen more benchmarks (Yang et al., 2024), and the Open LLM Leaderboard ranks pretrained models on six benchmarks, MMLU-Pro and MATH among them (Fourrier et al., 2024). Whether these scores say which base model will be best once fine-tuned is rarely tested. The closest test is that of Zeng et al. (Zeng et al., 2025), who fine-tune fifty 1B models pre-trained with varied objectives and data and find that perplexity on the Pile development set orders pairs of the fine-tuned systems worse than chance (pairwise accuracy 0.33–0.38); among their unsupervised indicators, few-shot accuracy on the target task does best. The received view is therefore that loss does not predict fine-tuning. We measure loss on the target domain's own text, across families and sizes, and it predicts the fine-tuned order closely; text from another domain predicts it less well (Section 5.4).

Loss, compression and capability. Language modelling was grounded in the entropy of text from the start (Shannon, 1951). Token-level perplexity depends on each model's vocabulary (Mielke, 2019; Ahia et al., 2023); normalising by UTF-8 bytes restores a common unit (Gao et al., 2021; Magnusson et al., 2023), and loss-based metrics carry more signal and less noise than accuracy across hundreds of models (Heineman et al., 2025). Language modelling is equivalent to compression (Delétang et al., 2024), and compression of fresh text correlates linearly with benchmark scores across about thirty models (Huang et al., 2024). UncheatableEval (UncheatableEval contributors, 2024) maintains a leaderboard of compression on recently published text for the same reason we use fresh text. These works check loss against benchmarks; we check it against fine-tuned systems, and find it predicts them at least as well as the closest benchmark and far better than GSM8K and MMLU-Pro.

How loss is computed. The Hugging Face guide to perplexity joins a test set into one stream and scores it with a sliding window (Hugging Face, 2024); lm-evaluation-harness scores each document separately, from a beginning-of-sequence token (Gao et al., 2024). Both are in common use, and the choice is treated as a detail. We show that it reorders models, and which choice predicts fine-tuning.

Choosing a model to fine-tune. Lin et al. predict which LLM will fine-tune best by fitting a rectified scaling law to small fine-tuning runs on the target task (Lin et al., 2024); this needs task data and training, where our measurement needs neither. Transferability estimators such as LEEP and LogME (Nguyen et al., 2020; You et al., 2021) score a pre-trained model from its features on labelled target data. Train-before-test (Zhang et al., 2025a) gives 61 models identical supervised preparation across 24 tasks and raises cross-benchmark agreement. Observational scaling laws relate benchmark performance across model families to a few capability measures (Ruan et al., 2024), and loss predicts downstream error within controlled families (Gadre et al., 2025). Thrush et al. use the varying correlation between per-domain perplexity and benchmark scores to select pre-training data (Thrush et al., 2024). Domain-adaptive pre-training (Gururangan et al.,

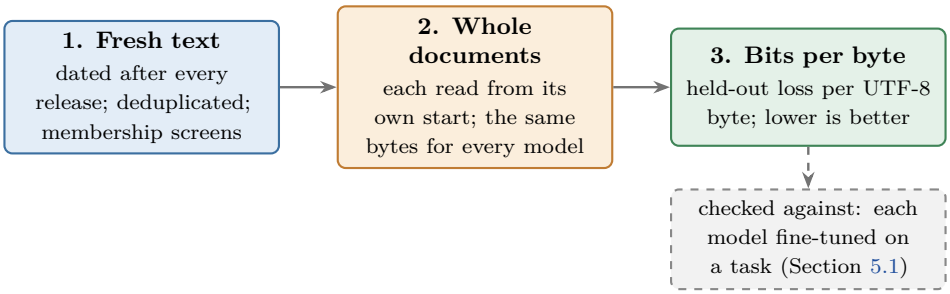


Figure 2: The method. The corpus is fixed and screened first; every candidate then reads the held-out documents whole and is scored in bits per byte. The fine-tuning experiments of Section 5.1 are not part of the method; we use them only to check its predictions.

2020) and dynamic evaluation (Krause et al., 2018) adapt a model before measuring it; for ranking, we find this unnecessary once documents are read whole.

Contamination. Models memorise duplicated pre-training text (Carlini et al., 2021; 2023), so deduplication and contamination analysis matter (Lee et al., 2022; Xu et al., 2024). Reference-free membership signals (Shi et al., 2024; Zhang et al., 2025b) are weak for large language models (Duan et al., 2024), which is why our main protection is the date of the text and the screens serve only for conservative removal. Fresh benchmarks such as LiveBench (White et al., 2025) apply the same idea to labelled questions.

3 Method

The method has three phases (Figure 2). The corpus is collected and screened before any model is run on it, every candidate then reads the held-out documents whole, and its loss is counted in bits per byte. There is no training and no label at any point.

Phase 1: a decontaminated corpus. The evaluation text must be text that no candidate has seen, and we use three safeguards for this. *Time:* Reddit and Hacker News were scraped in August 2026, after every model in the cohort was released, and the arXiv titles and abstracts were posted between 9 June and 20 August, after most of the releases. *Hygiene:* we remove exact duplicates, text that fails the Gopher quality heuristics (Rae et al., 2021) and MinHash/LSH near-duplicates (Broder, 1997) at Jaccard 0.60. The 80/10/10 split shares no identifier or normalised text across splits. *Membership screens:* the news snapshot (Google News, 8 June 2026, 119,054 articles) predates some of the releases, so we also test every model for having seen it. Min-K%++ (Zhang et al., 2025b) flags passages whose least likely tokens are unusually likely for the model, and a context-sensitivity test flags passages whose loss drops more than expected when the model is shown other text from the same corpus. We remove every passage flagged for any model, 80 in total, so all candidates are measured on the same corpus (Appendix C). The corpora, their manifests and record fingerprints are released.[1]

Phase 2: whole documents. Each test document is read on its own, starting from its tokenizer's beginning-of-document token, if it adds one, with up to 2,048 tokens of context. A document longer than that is read in overlapping windows with a stride of 1,024, and each token is scored once. The beginning-of-document token is never scored. On each corpus every model reads the same test documents, up to 6.0 MB, so the text is identical byte for byte. The implementation is released.[2] The per-model scores are released with the analysis that turns them into the numbers below.[3]

[1] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/docs/DATASET.md
[2] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/lm_adapt_bench/evaluate.py
[3] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/results/doc_protocol_cells

Phase 3: bits per byte. Let $\mathcal{L}$ be the total cross-entropy, in nats, over the scored tokens of the test text, and $|B|$ the UTF-8 byte length of the text those tokens cover:

$$\mathrm{BPB} = \frac{\mathcal{L}}{|B| \ln 2}. \tag{2}$$

A model with a larger vocabulary predicts fewer, longer tokens; dividing by bytes rather than tokens removes the penalty per-token perplexity would give it.

4 Experimental Setup

Models. Eleven base checkpoints between 0.5B and 35B parameters are evaluated on all four corpora, on blocks as well as whole documents (Table 9). Most are dense Transformers; one is a sparse mixture-of-experts, and LiquidAI's LFM2.5-1.2B is a hybrid of convolution and attention blocks. On news and on arXiv mathematics we add six more models, chosen before they were run to fill gaps in model size and to add publishers, for 17 in total (Table 9); their whole documents are also scored on Reddit and Hacker News. They come from the Llama, Qwen, Gemma, Ministral, Mistral-NeMo, Falcon, SmolLM and LFM families (Dubey et al., 2024; Meta AI, 2024; Yang et al., 2024; Qwen Team, 2026; Gemma Team, 2026; Liu et al., 2026; Mistral AI, 2024; Technology Innovation Institute, 2024; Ben Allal et al., 2025; Liquid AI, 2026). Eight of the seventeen are Qwen models.

Corpora. We use four corpora: news, Reddit, Hacker News and arXiv mathematics. Table 4 gives their sizes and how each was screened.

Benchmarks. For comparison, we score every model on MMLU-Pro (a stratified 1,000-question subset, so one question is 0.1 points), HellaSwag (all 10,042 validation items) and GSM8K (all 1,319 test items) with *lm-evaluation-harness* 0.4.12 (Gao et al., 2024), on one machine with one software build, following the protocol in Appendix F. We use scores we measured ourselves, not the values reported on model cards, and we publish the harness logs.

The fine-tuning task. We fine-tune fifteen of the seventeen models (Table 9), each starting from its base weights, to write an article's opening paragraph, its lede, from the rest of the article. Gemma-4-31B and Qwen-3.5-35B-MoE are left out because the fine-tune's 1,280-token sequences and generation pass do not fit them on one GPU. All fifteen use one shared LoRA recipe (Appendix D) and are scored by ROUGE-L, an F-measure based on the longest common subsequence with the reference (Lin, 2004), on 500 held-out articles that are not part of the BPB test set. The pairs to compare and the statistic were fixed in code before the fine-tuning jobs ran (Appendix D). We chose this task because every article provides its own reference, the lede written by the journalist, so no annotation is needed, and writing a good lede requires reading the whole article and writing in the style of the domain.

Statistics. We compare rankings mainly by *pairwise accuracy*, the fraction of model pairs that a ranking orders in the same way as a reference. Pairwise accuracy counts only *size-matched* pairs, two models whose parameter counts differ by less than a factor of two; when one model is more than twice the size of the other, the larger one usually wins, and a ranking would get the pair right by following size. Because pairs that share a model are not independent, 95% intervals come from a cluster bootstrap over models (20,000 draws). Intervals for a single fine-tuned pair resample its 500 evaluation articles. With fifteen models the intervals are wide, and we say so where it matters.

5 Results

5.1 BPB on the domain's documents anticipates the fine-tuned ranking

A team will fine-tune whichever model it chooses, so the question is whether a measurement taken before any task exists can say how that fine-tuning will turn out. Measured with no task, no labels and no training,

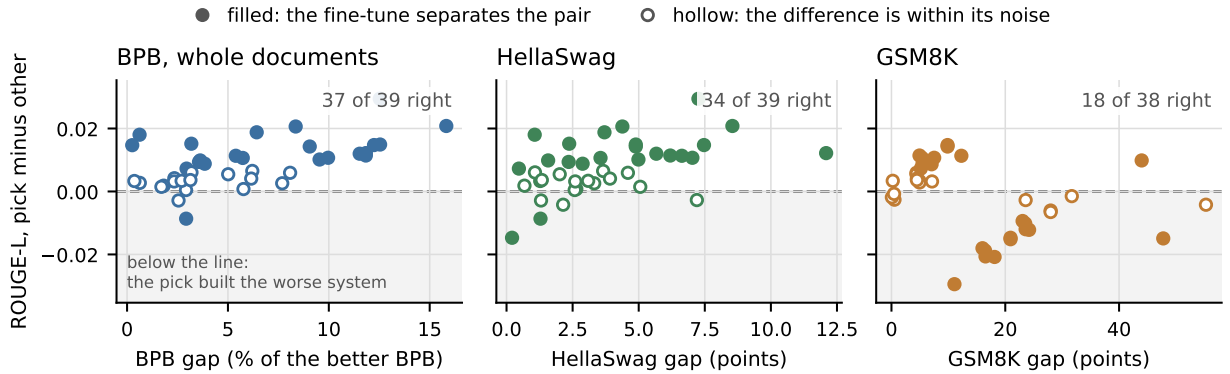


Figure 3: How well does each ranking predict the fine-tuned results? Each point is a pair of fine-tuned models whose parameter counts are within a factor of two (39 pairs; 38 for GSM8K, where two models tie). For each pair, x is how strongly the ranking prefers one model (the gap in that measure) and y is how much better that model's fine-tuned system scores in ROUGE-L. Points above the line are pairs the ranking orders correctly. Filled markers are *resolved* pairs, whose ROUGE-L difference has a 95% bootstrap interval over the 500 articles that excludes zero. BPB on whole documents gets 37 of 39 pairs right, including 21 of the 22 resolved pairs; HellaSwag gets 34; GSM8K gets 18 of 38, and all eleven of its pairs with gaps between 15 and 25 points fall below the line.

BPB on whole news documents orders 37 of the 39 size-matched pairs (0.949, 95% interval $[0.817, 1.000]$) in the same way as the fine-tuned systems, and 21 of the 22 pairs that the fine-tuning results themselves resolve (Figure 3, Table 1). HellaSwag orders 34 (0.872, $[0.640, 1.000]$); the difference, $+0.077$ $[0.000, 0.300]$, does not separate the two. GSM8K and MMLU-Pro are at chance, with 0.474 and 0.513. So is the strongest label-free baseline, the models' own scores on the task before any training (the few-shot baseline that Zeng et al. found best): it orders 20 of the 39 pairs and agrees with the fine-tuned order at Spearman 0.53. Fine-tuning reorders the models, and the document reading predicts the order they end in. Ranking by size agrees with the finish at Spearman 0.92 over the fifteen, because large models beat small ones, but orders 28 of the 39 size-matched pairs; BPB beats it by $+0.231$ in pairwise accuracy $[+0.059, +0.452]$. Over all fifteen models the document ranking agrees with the final order at Spearman 0.99: every model is within one rank of its final position, and eleven are exactly in place. GSM8K reaches 0.59, MMLU-Pro 0.71 and HellaSwag 0.97. On the 32 pairs whose models come from different publishers, document BPB orders 30 and HellaSwag 27.

Table 2 shows the models the benchmarks misjudge most. Gemma-4-12B, ninth on GSM8K and eighth on MMLU-Pro among the fifteen, produces the best fine-tuned system; in the pair fixed in advance it beats Ministral-3-14B by $+0.018$ ROUGE-L $[+0.011, +0.025]$. The 55.3-point GSM8K gap between Llama-3.2-1B and Qwen-2.5-1.5B did not separate their fine-tuned systems: that pair was inconclusive. Every model improved on its own base weights and beat the score of copying the article's first body sentence (ROUGE-L 0.125).

The prediction does not depend on the fine-tuning recipe. With a deliberately different recipe, recipe B (LoRA rank 64 and α=128, three times the learning rate, half the batch size, 400 steps and a new seed), the order of the fine-tuned systems hardly changes (Spearman 0.982), and document BPB is still the best predictor at 0.923, ordering correctly all 20 pairs that recipe B separates; its margin over size no longer resolves ($+0.179$ $[-0.036, +0.441]$).

5.2 Blocks cut from a stream misrank strong models

The same news text, scored the common way, gives a different ranking. Cut into non-overlapping 512-token blocks from the concatenated test set, each block read with no earlier context, it orders 30 of the 39 pairs (0.769) and agrees with the fine-tuned order at Spearman only 0.71. Gemma-4-12B, second on whole documents, is thirteenth of fifteen on blocks, behind models with fewer than 2B parameters, and LFM2.5-1.2B is last (Table 2).

	All 39 pairs	95% CI	22 resolved pairs	Recipe B
Adapted BPB, documents	0.974	[0.862, 1.000]	1.000	0.897
Unadapted BPB, documents	**0.949**	[0.817, 1.000]	0.955	0.923
Adapted BPB, blocks	0.949	[0.817, 1.000]	0.955	0.923
HellaSwag	0.872	[0.640, 1.000]	0.909	0.846
Unadapted BPB, documents, Reddit text	0.872	[0.659, 1.000]	0.909	0.897
Unadapted BPB, documents, Hacker News text	0.795	[0.481, 1.000]	0.818	0.769
Unadapted BPB, blocks	0.769	[0.444, 1.000]	0.682	0.744
Parameter count	0.718	[0.463, 0.912]	0.773	0.744
Unadapted BPB, documents, arXiv maths text	0.590	[0.260, 0.929]	0.545	0.615
Base weights scored on the task itself	0.513	[0.233, 0.842]	0.636	0.538
MMLU-Pro	0.513	[0.250, 0.846]	0.500	0.538
GSM8K	0.474	[0.194, 0.824]	0.455	0.447

Table 1: Which ranking predicts the fine-tuned systems? Fifteen models, each fine-tuned with the same recipe on the lede task and scored by ROUGE-L on 500 held-out articles. Each row gives the pairwise accuracy of one ranking over all size-matched pairs, with a bootstrap interval over models, and its accuracy on the 22 pairs whose fine-tuned difference is resolved by a bootstrap over articles. The last column scores the same rankings against systems trained with recipe B. BPB rows use news text unless marked, and the bold row is the paper's reading; the block and adapted readings are defined in Appendix A. "Base weights scored on the task itself" is each model's own ROUGE-L on the task before any training. GSM8K ties on one pair and is scored on 38.

	Before fine-tuning			After	BPB, documents
Model	BPB, blocks	GSM8K	MMLU-Pro	**fine-tuning**	**predicted**
Gemma-4-12B	13	9	8	**1**	2
Llama-3.2-1B	10	15	15	**11**	11
LFM2.5-1.2B	15	12	13	**14**	14

Table 2: Three models the benchmarks misjudge. Rank among the fifteen fine-tuned models (1 is best). Gemma-4-12B is mid-table on both benchmarks and near the bottom on block BPB, then builds the best system of all; Llama-3.2-1B is last on both benchmarks and finishes eleventh. BPB on whole documents places every model within a single rank of where it finishes: all fifteen, and eleven exactly.

Figure 4 shows why. Gemma-4-12B starts a block at 2.83 bits per byte, while Qwen-2.5-1.5B, eight times smaller, starts at 2.04. Gemma stays behind for most of the block and is ahead by the end, 0.72 against 0.77. The early positions have the highest loss, so they contribute most of the difference between the two block averages. Blocks are cut from a packed stream, so most start in the middle of a document, without the beginning-of-document token that models such as Gemma are trained to see first.

Two tests, with their decision rules fixed before they ran, point at the cut rather than the model. Scoring only positions 64, 128 or 256 onwards of each block removes the first tokens of the block but not the mid-document start, and on the 10 size-matched fine-tuned pairs whose models we scored this way it leaves the block reading at 6 correct, unchanged. Reading every test document whole, from its own start, ranks the fifteen models as in Section 5.1, and Gemma-4-12B moves from thirteenth to second. The document reading is how lm-evaluation-harness scores text, with no parameter to tune; we added it after the fine-tuning results were known, with its rule fixed before it ran. We did not separate the beginning-of-document token, the document boundary and the longer context; together, they amount to reading the text as the model was trained to see it.

5.3 No training is needed

Adapting every model briefly to the domain before scoring it is a natural precaution, and on blocks it is a repair: adapted block BPB ranks the fifteen models exactly as whole documents do, because the models that gain most from adaptation are the ones blocks misrank (Appendix A). On whole documents it changes little. Adapting each model with the same recipe and then reading the documents lowers every model's news loss,

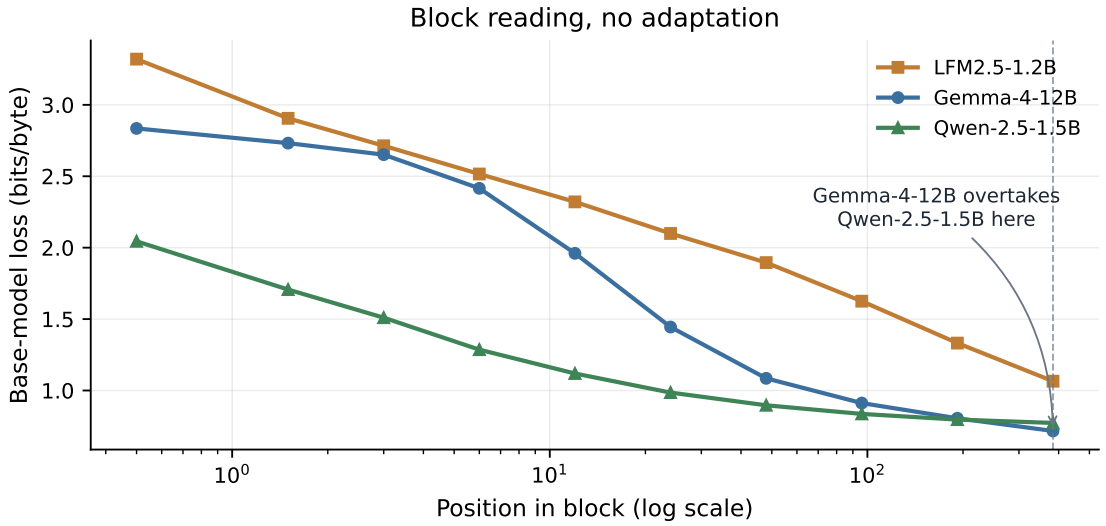


Figure 4: The start of each block costs some models far more than others. Loss by position within a 512-token evaluation block, for Gemma-4-12B, Qwen-2.5-1.5B and LFM2.5-1.2B. Gemma-4-12B starts a block at 2.83 bits per byte, against 2.04 for Qwen-2.5-1.5B, and ends it at 0.72 against 0.77.

	Rank correlation (ρ) with		
Text scored	GSM8K	MMLU-Pro	HellaSwag
News	0.755	0.827	0.991
Reddit	0.691	0.764	0.973
Hacker News	0.782	0.873	0.982
Math (arXiv)	0.918	0.945	0.836

Table 3: Which benchmark the document ranking agrees with depends on the text. Rank correlation between BPB on whole documents and each benchmark, over the eleven original models; ρ is signed so that 1 means lower BPB goes with a higher score. On the three general corpora the ranking is closest to HellaSwag; on arXiv mathematics it is closer to GSM8K and MMLU-Pro. The rule was fixed before the document reading of the three new corpora ran.

by 2.0–7.9%, but the two document rankings agree at Spearman 0.993: two neighbouring pairs swap. In the one that is size-matched, Gemma-4-12B, which gains the most, moves past Mistral-Nemo-12B into first place; the adapted documents order 38 of the 39 pairs, one more than the unadapted reading, inside the same interval. The same holds on the other corpora: for the eleven original models, the adapted and unadapted document rankings agree at Spearman 0.982 on Reddit, on Hacker News and on arXiv mathematics, with adaptation lowering loss by 1.9–11.2%. We fixed the threshold for this claim, 0.95, before the adapted runs.

5.4 The text has to be like the task's

The document reading predicts the news fine-tune when it reads news. Reading arXiv mathematics instead, the same fifteen models are ordered correctly on 23 of the 39 pairs (0.590, $[0.260, 0.929]$) and on 12 of the 22 resolved pairs, below ranking by model size (Table 1). Text close to news does better. Reddit and Hacker News are general prose like news, and on the same fifteen models Reddit orders 34 of the 39 pairs (0.872, $[0.659, 1.000]$) and Hacker News 31 (0.795, $[0.481, 1.000]$). The order is news 37, Reddit 34, Hacker News 31, model size 28, mathematics 23. The three general corpora also rank the eleven original models almost alike (Spearman 0.982 and 0.973 with news; mathematics 0.818; Table 8).

Which benchmark the ranking agrees with also follows the text (Table 3). On news, Reddit and Hacker News, document BPB agrees with HellaSwag at Spearman 0.97–0.99, and with GSM8K at 0.69–0.78. On arXiv mathematics the order reverses: 0.92 with GSM8K, 0.95 with MMLU-Pro and 0.84 with HellaSwag.

We wrote this rule down before reading the three new corpora as whole documents; the block readings had already shown the same pattern (Appendix A). HellaSwag is close on news; on mathematics text the ranking follows GSM8K and MMLU-Pro instead, so no one benchmark stands in for the domain's own text.

5.5 The measurement is deterministic

The document reading involves no randomness: the adapted jobs re-read every unadapted model and reproduced its document BPB exactly, and the control models reproduce the published unadapted block readings on a second host to within 0.04%.

BPB has a structural advantage over generative benchmarks here. Floating-point addition is not associative, so results on a GPU can change with batch size and kernel choice (He & Thinking Machines Lab, 2025). A generative benchmark decodes greedily, so a tiny change in two nearly tied logits can flip a token, and the model then continues from its own changed output. BPB is computed with teacher forcing: every prediction is conditioned on the true text, so a small perturbation changes one log-probability slightly and does not propagate. Generative scores are also sensitive to decoding settings that BPB does not use: dropping a single decoding setting shipped with Gemma-4-12B changed its GSM8K score by 9.86 points and its MMLU-Pro score by 14.30 (Appendix F).

6 Limitations

The fine-tuned evidence is one task, lede writing in news, with two LoRA recipes at a small fixed budget and fifteen models, seven of them Qwen, so the intervals are wide and BPB and HellaSwag are not separated. We piloted two fine-tuning tasks on arXiv mathematics, writing a title from its abstract and the reverse; on both, a 0.5B and a 12B model finished within 0.012 ROUGE-L of each other, so by a rule fixed in advance neither task could rank models and neither was run further. The two largest models were not fine-tuned. Membership screens cannot prove that a proprietary training set excludes a text; the dates of the Reddit, Hacker News and arXiv corpora are the stronger guarantee.

7 Conclusion

Base models are chosen by benchmark scores, but those scores grade abilities the task may not use. We ranked candidates instead by their loss, in bits per byte, on fresh text from the target domain, with each document read whole from its own start. It needs no labels, no task definition and no training, only a forward pass over a few megabytes of text.

On a news-writing task this ranking placed all fifteen fine-tuned systems within one rank of where they finished and ordered 37 of 39 size-matched pairs correctly. GSM8K and MMLU-Pro were at chance, as were the models' own scores on the task before training; HellaSwag ordered 34. Two choices decide whether the measurement works. Text from another domain, arXiv mathematics, ordered only 23 of the pairs. Text cut into fixed-length blocks put Gemma-4-12B, the model that fine-tunes best, thirteenth of fifteen. With the domain's own documents read whole, adapting the models first changed almost nothing. A new corpus can be collected for every domain and every evaluation, so the text the measurement reads stays unseen.

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Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K%++ / context / perturbation union
Reddit, 6–10 Aug 2026[†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv `math.*`, 9 Jun–20 Aug 2026[‡]	3,207	3,207	–	15 MB	Post-release scrape + GSM8K shingle screen

Table 4: Corpora. All four receive identical exact deduplication, Gopher quality filtering and MinHash/LSH near-duplicate removal at Jaccard 0.60; they differ in contamination control. The membership screens ran on the news corpus, whose 8 June 2026 snapshot predates part of the cohort's release window; Reddit and Hacker News were scraped after every model shipped, as was most of the arXiv set. Reddit comments are concatenated within a subreddit, and Hacker News items in sequence, into documents of about 600 words, which is why they have fewer documents than raw records. [†]Four days are available in the stated range. [‡]Used for the mathematics adaptation.

A The Block and Adapted Readings

Sections 5.2 and 5.3 compare the document reading with these.

Blocks, unadapted and adapted. Every candidate is adapted to the retained training text, packed into contiguous 512-token blocks, using LoRA (Hu et al., 2022) on all linear layers, and then scored on non-overlapping 512-token test blocks cut from the packed test stream, each read with no earlier context. Beginning-of-document tokens inside a block are used as context but never scored. Because blocks are cut in tokens, each model covers a slightly different amount of text, 5–10% apart (Appendix C). The same block reading without adaptation is the *unadapted block* reading. For adaptation, each candidate can get its own hyperparameter search with Optuna (Akiba et al., 2019). A Tree-structured Parzen Estimator (TPE; Bergstra et al., 2011) proposes the learning rate, batch size, warm-up, weight decay, LoRA rank, α and dropout; successive halving (Jamieson & Talwalkar, 2016) prunes weak trials from 50 steps on; and every trial is scored by BPB on the validation split, never on the test split. Every model gets the same number of trials and no time limit, because with a wall-clock limit small models complete more trials than large ones and end up better tuned (Appendix C). The best configuration is then trained until validation BPB stops improving. The released pipeline implements this procedure.[4] Every adapted number in this paper uses a single shared configuration instead, for compute reasons, described next. Adapted models can also read whole documents (Section 5.3).

Adaptation runs. Because of limited compute, all runs reported here use one shared configuration instead of the per-model search: LoRA rank 16, α=32, dropout 0.05, learning rate 10^{-4}, effective batch size 32 and 250 update steps, a common LoRA setting. The runs are a fixed budget, not trained to convergence. This is 4.1M training tokens per candidate and costs 3.7 GPU-minutes on one A100 at 0.5B parameters, 11.6 at 7.7B, 24.7 at 12B and 62.7 at 31B; at 512 tokens per sequence every model's adaptation fits on one 64 GB GPU. The budget is matched in tokens, so models with larger vocabularies see a few percent more bytes of text.

What adaptation does to the block reading. For the eleven models, the block rankings before and after adaptation have a Spearman correlation of only 0.655 on news, 0.609 on Reddit and 0.682 on Hacker News, and one model moves seven places (Table 5). The changes are uneven. On news, adaptation lowers

[4] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/lm_adapt_bench

Model	Params (B)	News BPB, blocks (rank)		Moves	Gain	Benchmark score, % (rank)		
		unadapted	adapted			HellaSwag	GSM8K	MMLU-Pro
Gemma-4-31B	31.4	0.752 (4)	0.624 (1)	↑**3**	**17.0%**	87.9 (1)	85.0 (3)	57.1 (2)
Gemma-4-12B	12.0	0.866 (9)	0.653 (2)	↑**7**	**24.5%**	84.0 (3)	64.8 (7)	44.8 (7)
Qwen-3.5-35B-MoE*	34.7	0.707 (1)	0.678 (3)	↓2	4.1%	85.6 (2)	86.6 (1)	61.4 (1)
Ministral-3-14B	14.0	0.718 (2)	0.679 (4)	↓2	5.4%	82.9 (4)	80.8 (6)	52.8 (4)
Qwen-3.5-9B	9.0	0.740 (3)	0.708 (5)	↓2	4.4%	81.6 (5)	85.8 (2)	55.6 (3)
Qwen-2.5-7B	7.7	0.758 (5)	0.717 (6)	↓1	5.4%	80.3 (6)	81.3 (4)	47.2 (6)
Qwen-3.5-4B	4.2	0.782 (6)	0.749 (7)	↓1	4.2%	77.2 (7)	81.0 (5)	52.4 (5)
Llama-3.2-1B	1.2	0.811 (7)	0.771 (8)	↓1	4.9%	65.8 (9)	6.4 (11)	11.2 (11)
Qwen-2.5-1.5B	1.6	0.833 (8)	0.797 (9)	↓1	4.3%	68.0 (8)	61.6 (8)	25.6 (8)
LFM2.5-1.2B	1.2	1.340 (11)	0.852 (10)	↑1	**36.4%**	60.9 (10)	54.1 (9)	16.4 (9)
Qwen-2.5-0.5B	0.5	0.935 (10)	0.901 (11)	↓1	3.6%	51.8 (11)	34.3 (10)	16.0 (10)

Table 5: The eleven models on the block readings. Sorted by adapted news block BPB (lower is better). *Moves* is the number of places a model gains or loses with adaptation, and *Gain* is the reduction in loss. Three models gain far more than the rest (the two Gemmas and LFM2.5-1.2B), and they are the only ones that move up. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters.

block loss by 3.6–5.4% for every Qwen, Llama and Ministral model, but by 17.0% for Gemma-4-31B, 24.5% for Gemma-4-12B and 36.4% for LFM2.5-1.2B. These three are the only models that move up, and they are the models the block reading misranks. For the fifteen fine-tuned models, adapted block BPB ranks them exactly as the unadapted document reading does (Spearman 1.00). Most of what adaptation did to the block reading was teaching the models to read text that starts mid-document.

Adapted blocks and the benchmarks. Before adaptation, block BPB on news, Reddit and Hacker News has a Spearman correlation of 0.63–0.72 with HellaSwag across the eleven models; after adaptation it is 0.96–0.98, close to the document reading's 0.97–0.99. Adapted on arXiv mathematics instead, block agreement rises with all three benchmarks: with GSM8K from 0.76 to 0.91, with MMLU-Pro from 0.74 to 0.93, and with HellaSwag from 0.48 to 0.82 (Table 6).

Seeds. The adapted readings depend on the adaptation's seed, so we ran three seed replicates of all 17 news adaptations; the seed sets the adapter's initialisation, dropout and batch order, and the data splits and test set stay fixed. None of the 40 size-matched pairs changes order across seeds. The median standard deviation of adapted block BPB within a model is 0.0001, against 0.074 between models; the largest gap seed noise could open between two runs is 1.24% of the median BPB ($2\sqrt{2}$ standard deviations of the noisiest model, Gemma-4-12B; the median model, under 0.04%).

B The Full Benchmark Alignment

Table 6 reports all forty-eight (corpus × reading × benchmark) cells over the full 55 pairs, and Table 7 gives, for each reading, how many benchmark points its top-ranked model gives up against each benchmark's own top model. On blocks, adapting on general text raises HellaSwag agreement on every corpus and lowers GSM8K agreement on every corpus and MMLU-Pro agreement on two of the three. Adapting on mathematics raises all three. Rank correlations hide how close these are in pair counts: across its sixteen rows the GSM8K column spans only 41 to 50 of 55 pairs. Adapting barely moves the document reading: on the three general corpora its adapted rows equal the adapted block rows.

C Robustness

Consistency across domains. All eleven models ran on three independently collected general-text corpora (Tables 4 and 10). Every reading ranks them almost identically across these corpora. On whole documents news and Reddit agree at $\rho = 0.982$, news and Hacker News at 0.973 and Reddit and Hacker News at 0.964. The block readings are as consistent, before adaptation (0.982, 0.964, 0.973) and after (0.991, 0.955, 0.964), although on each corpus adaptation reorders the blocks (Spearman 0.655 on news, 0.609 on Reddit and 0.682

Corpus	Reading	Rank correlation (ρ) with			Pairs ordered the same way, of 55		
		GSM8K	MMLU-Pro	HellaSwag	GSM8K	MMLU-Pro	HellaSwag
News	zero-shot	0.800	0.845	0.718	45	45	44
	adapted	0.700	0.773	0.982	42	44	53
	document	0.755	0.827	0.991	43	45	54
	document-adapted	0.700	0.773	0.982	42	44	53
Reddit	zero-shot	0.745	0.782	0.655	43	43	42
	adapted	0.655	0.745	0.964	41	43	52
	document	0.691	0.764	0.973	41	43	52
	document-adapted	0.655	0.745	0.964	41	43	52
Hacker News	zero-shot	0.800	0.809	0.627	44	44	43
	adapted	0.745	0.855	0.973	44	46	53
	document	0.782	0.873	0.982	44	46	53
	document-adapted	0.745	0.855	0.973	44	46	53
Math (arXiv)	zero-shot	0.764	0.736	0.482	43	43	40
	adapted	0.909	0.927	0.818	49	49	46
	document	0.918	0.945	0.836	48	50	47
	document-adapted	0.936	0.945	0.845	50	50	47

Table 6: Agreement of each BPB reading with each benchmark. Rank correlation of BPB with each benchmark over the eleven models, before (zero-shot) and after adaptation, on blocks and on whole documents, on every corpus; ρ is signed so that 1 means lower BPB goes with a higher score. On the three general corpora adapting the block reading lifts HellaSwag from 0.63–0.72 to 0.96–0.98 while GSM8K falls; on arXiv mathematics it lifts all three, and GSM8K and MMLU-Pro end above HellaSwag. The last three columns count, of all 55 pairs of the eleven (not only size-matched ones), those that BPB orders the same way as each benchmark: adapted news BPB agrees with HellaSwag on 53.

on Hacker News between the two block rankings, Appendix A). The adapted block values agree as well as the ranks (Figure 5).

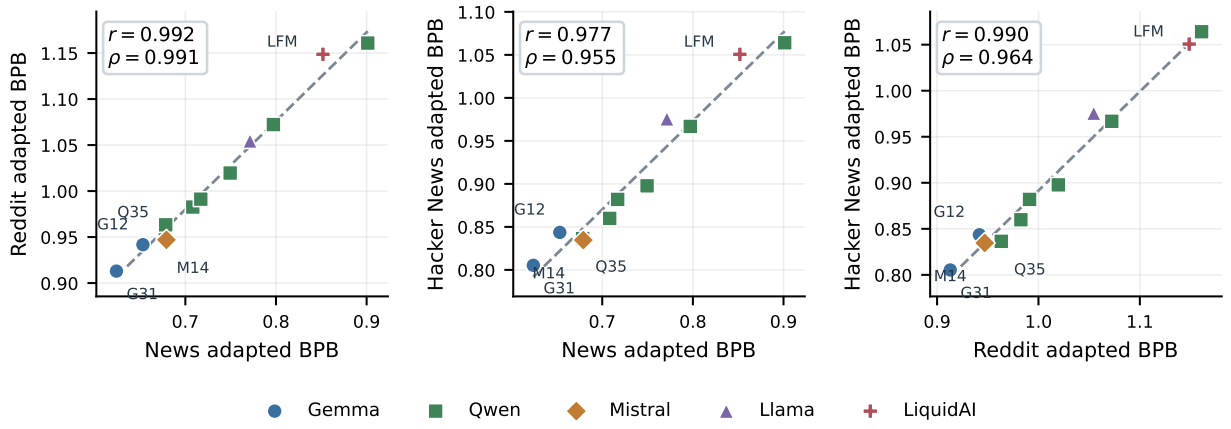


Figure 5: Adapted block BPB across domains. Each point is one model, its adapted BPB on two corpora. Absolute difficulty changes, but points remain close to a monotone shared ordering: Pearson correlations between the values are 0.992 (news–Reddit), 0.977 (news–Hacker News) and 0.990 (Reddit–Hacker News). G31/G12 denote Gemma-4-31B/12B, Q35 Qwen-3.5-35B-MoE, M14 Ministral-3-14B and LFM LFM2.5-1.2B.

Why the search gives every model the same number of trials. The per-model search (Appendix A) uses a multivariate TPE sampler (seed 42) over learning rate (10^{-6}–10^{-3}, log), physical batch (2, 4, 8), gradient accumulation (1–16), warm-up ratio (0–0.2), weight decay (10^{-5}–10^{-1}, log), LoRA rank (8–64), α as a multiple of rank (1, 2, 4) and dropout (0–0.2); a successive-halving pruner starts trials at 50 steps and

Corpus and reading	Picks	Accuracy given up (points)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	1.6	4.3	0.0
News document	Gemma-4-31B	1.6	4.3	0.0
News document-adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit zero-shot	Ministral-3-14B	5.8	8.6	5.0
Reddit adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit document	Gemma-4-31B	1.6	4.3	0.0
Reddit document-adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News document	Gemma-4-31B	1.6	4.3	0.0
Hacker News document-adapted	Gemma-4-31B	1.6	4.3	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) document	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) document-adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 7: The cost of the top-1 pick on each benchmark. For every corpus and reading, the model the ranking puts first and how many accuracy points that model is below each benchmark's own top model. Only three distinct picks occur in sixteen rows, and both document readings pick the same model as adapted block BPB on every corpus. Adapted BPB on every general-text corpus picks Gemma-4-31B, the HellaSwag leader, at 1.6 GSM8K and 4.3 MMLU-Pro points. Adapted mathematics BPB picks Qwen-3.5-35B-MoE, the GSM8K and MMLU-Pro leader, which unadapted block BPB on news, Hacker News and arXiv also picks.

Corpora compared	**Rank agreement** (Spearman)			**Value agreement** (Pearson)	
	documents	blocks	adapted blocks	blocks	adapted blocks
News–Reddit	0.982	0.982	0.991	0.996	0.992
News–Hacker News	0.973	0.964	0.955	0.994	0.977
News–arXiv maths	0.818	0.909	0.755	0.964	0.839
Reddit–Hacker News	0.964	0.973	0.964	0.998	0.990
Reddit–arXiv maths	0.764	0.918	0.736	0.971	0.882
Hacker News–arXiv maths	0.882	0.973	0.836	0.982	0.930

Table 8: Agreement between corpora. Agreement between the eleven models' BPB on two corpora. The three general-text corpora rank the models almost identically under every reading (0.96–0.98 on whole documents). Mathematics agrees less with the general corpora (0.76–0.88 on documents), most of all after adaptation on blocks.

promotes the best third at each rung; and the objective is BPB on a fixed 256-block subset of the validation split. An earlier run of this search on the eleven-model news cohort was capped by wall-clock time rather than by trial count, and the cap was confounded with model size: the two largest models completed four trials each while the 1B models completed 17 and 18 (`results/legacy_sweep/*/trials.csv`), and the five models of 9B or more all kept the same third draw of the seeded sampler as their winner. A time-capped search hands small models a better-tuned adapter and biases the ranking by parameter count, which is why the released pipeline budgets the search in trials and applies no time cap by default. Apart from the right panel of Figure 4, none of those adapters produces a number in the paper.

Membership screens. The news corpus passes three reference-free membership screens per model: Min-K%++ (normalised likelihood of the lowest-probability K=20% target tokens), context sensitivity (target loss with and without sampled same-corpus prefixes) and, for flagged passages only, preference for the original passage over two word-swap perturbations. The union over models and screens removed 80 unique sequences, after which every model receives the identical retained corpus. Near-100% preference for the original is expected from fluency alone, so we draw no memorisation conclusion from the third test (`results/contamination/`).

Context length. Varying the inference context on 64 fixed held-out target windows from 16 to 2048 tokens, both models measured improve with context and the larger improves faster, so a protocol that scores only short contexts under-rates models that use long ones. Measured context sensitivity is 0.032 bits per byte per doubling, the constant the bound below uses. The document reading of Section 3 scores up to 2,048 tokens of context.

Token-aligned blocks: a small bias, bounded. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text: 5.92–6.24 MB across the eleven-model cohort, a 5% spread, and 9.9% across the seventeen. Bounding the resulting bias with the measured context sensitivity puts it at 0.08× the smallest adjacent rank gap in the eleven-model cohort. In the extended cohort the worst ratio of bound to gap is 0.61, for Qwen2.5-3B over Falcon3-7B; the bound overturns no rank. The document reading covers identical bytes for every model and removes the question.

Seed variation is negligible. The published adaptation cells did not fix the `torch` random seed, which sets the LoRA initialisation, dropout and batch order (the data splits are seeded separately). A check fixed before it ran re-ran all 17 news models with three explicit seeds (51 cells, everything else pinned). The median within-model standard deviation of adapted BPB is 0.0001 against a between-model standard deviation of 0.074, and none of the 40 pairs of models within 2× in size reverses its order across seeds. Gemma-4-12B is the one model where seed noise is visible (standard deviation 0.003, five times the next largest).

The mathematics shift. Adapting on general text raises agreement with HellaSwag on news from $\rho = 0.718$ to 0.982 while GSM8K falls from 0.800 to 0.700 and MMLU-Pro from 0.845 to 0.773; adapting on arXiv mathematics raises GSM8K agreement to 0.909 and recovers GSM8K's top model, while the three general-text corpora all leave GSM8K at or below its unadapted agreement. The shift predicts a *joint* movement, so we test it with a difference in differences: (GSM8K minus HellaSwag selection accuracy under mathematics adaptation) minus the same quantity under a general-text corpus, which is +0.242 pooled. Resampling models gives a strictly positive difference in 93.7% of draws (one-sided $p \approx 0.06$). Resampling whole publisher families, which the shared training data of a family calls for, gives 80.9% ($p \approx 0.19$), with ties counted as non-positive; not one of its 20,000 draws is negative. A leave-one-family-out jackknife shows the dependence: dropping the two Gemma models takes the pooled difference from +0.242 to +0.056. More families, rather than more models, would settle it.

Reproduction controls. Re-running two Ministral-3-14B cells weeks later on different hardware reproduced unadapted BPB exactly and adapted BPB to within 0.0003. Because the six added models were adapted on a different software stack, a published cell (Qwen2.5-7B) was re-run through that stack; the cohorts are pooled only because it reproduces within 1%, and it reproduces to 0.01% on adapted BPB (0.7169 against 0.7170).

D The Downstream Fine-Tune

Section 5.1 reports the result; this section describes how it was built. It ran in two stages, each fixed in code before its job returned: first the two pairs on which the rankings disagree most (Gemma-4-12B against Ministral-3-14B, Llama-3.2-1B against Qwen-2.5-1.5B), then every cohort model that fits one GPU, so that no pair is hand-picked. Gemma-4-31B and Qwen-3.5-35B-MoE need more than one GPU for this fine-tune, because its 1,280-token sequences and its generation pass cost far more memory than the 512-token adaptation (Appendix A), and are excluded by that rule, which costs one size-matched pair of forty. The fine-tuned scores, pairs and rankings are released.[5]

The task. Journalistic convention makes a news article's first paragraph a standalone summary of it, so the corpus supplies its own supervision: the input is the article with its opening paragraph removed, truncated to 3,000 characters, and the target is that paragraph. Articles qualify if they have at least three paragraphs, an opening paragraph of 200–700 characters ending in sentence punctuation, and at least 800 characters of

[5] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/results/downstream.json

remaining body. Of the 11,905 validation documents, 3,627 qualify; 2,000 are sampled for training and 500 for evaluation under a fixed seed.

Why the validation split. The BPB test blocks live in the *test* split, so the fine-tuning task is built from the validation split instead. Evaluation documents are hashed against the test documents twice, on full normalised text and on a 400-character opening, the second because a syndicated wire story is republished with an edited body and an identical lede. Four candidate articles were dropped by that second check, before the 500 were sampled, and none by the first.

Equal preparation. Every model gets the adaptation recipe unchanged (LoRA rank 16, α=32, dropout 0.05, learning rate 10^{-4}, effective batch 32, 250 steps), starting from its base weights, with prompt tokens masked out of the loss so the model is trained to write the lede rather than to reconstruct the article. The tokenizers differ, so the budget is matched in update steps $\times$ batch $\times$ sequence length; each model's realised token count is in the artifact. Decoding is greedy with a 180-token cap and each checkpoint's own generation configuration otherwise untouched, following Appendix F. The prompt is plain text: the article body followed by an instruction to write its opening paragraph. When the prompt is too long, the body is cut, never the instruction after it, because cutting the assembled prompt would delete the instruction from the longest articles, at a different article count for each tokenizer. At most four of the 500 evaluation articles were cut for any model, and none for three of them.

The decision rules, fixed before each run. For the two chosen pairs: mean ROUGE-L F1 over the 500 held-out articles, a paired bootstrap over articles (20,000 draws, 95% percentile interval) and a practical-equivalence margin of 0.01; an interval excluding zero resolves the pair for whichever ranking predicted that winner. The first pair resolved for adapted BPB; the second was inconclusive. For the full cohort the fine-tuned ROUGE-L is the outcome, and every ranking is scored by pairwise accuracy within 2$\times$ with a model-level bootstrap of 20,000 draws (Table 1). A separate guard voids any comparison in which fine-tuning failed to improve a model over its own base weights, since ordering unmoved models would order their base behaviour under another name. It did not fire: all fifteen models gained between +0.041 and +0.153.

A second recipe. Recipe B changes every setting the adaptation and the fine-tune shared (LoRA rank 64, α=128, learning rate 3×10^{-4}, effective batch 16, 400 steps, a new seed) on the same task, data, decoding and metric, and was fixed before it ran. It fits the training articles far harder and scores slightly below recipe A for every model, yet orders the fifteen systems almost identically (Spearman 0.982). Against recipe-B systems adapted block BPB, and document BPB with the same order, remain the top ranking at 0.923 $[0.750, 1.000]$, ahead of HellaSwag at 0.846, parameter count at 0.744 and GSM8K and MMLU-Pro near chance, and it picks the better system on all 20 pairs recipe B resolves. Its margin over parameter count, resolved under recipe A, does not resolve here ($+0.179$ $[-0.036, +0.441]$), so that margin is recipe-specific (Table 1).

What copying scores. A news body restates its own lede, so reference overlap is partly available by copying. The article's first body sentence scores 0.125 against the reference and a random body sentence 0.111. Unadapted base weights land close to copying (inspection confirms they largely do copy), while every fine-tuned system clears it, from 0.188 (Qwen-2.5-0.5B) to 0.249 (Gemma-4-12B).

How close the corpus has to be. Exploratory, run after the result: on the ten size-matched pairs of the nine fine-tuned models adapted on all four corpora, adapted BPB orders the news fine-tune with pairwise accuracy 0.900 for news and for Reddit, 0.700 for Hacker News and 0.600 for arXiv mathematics; the unadapted block readings reach 0.500–0.600. Ten pairs support no interval, so this is a direction, not a result. The unadapted document reading on all fifteen models and 39 pairs is in Table 1.

E The Cohorts

Table 9 is the 17-model cohort: every model, its parameter count, its adapted news BPB and the analyses it enters. Table 10 gives the original eleven across all three general corpora and both readings.

Model	Params (B)	Adapted BPB	Cohort
Gemma-4-31B	31.4	0.624	original
Mistral-Nemo-12B	12.3	0.646	**added**
Gemma-4-12B	12.0	0.653	original
Qwen-2.5-14B	14.8	0.675	**added**
Qwen-3.5-35B-MoE*	34.7	0.678	original
Ministral-3-14B	14.0	0.679	original
Qwen-3.5-9B	9.0	0.708	original
Qwen-2.5-7B	7.7	0.717	original
Falcon3-10B	10.4	0.728	**added**
Qwen-3.5-4B	4.2	0.749	original
Qwen-2.5-3B	3.1	0.756	**added**
Falcon3-7B	7.5	0.758	**added**
Llama-3.2-1B	1.2	0.771	original
SmolLM2-1.7B	1.7	0.788	**added**
Qwen-2.5-1.5B	1.6	0.797	original
LFM2.5-1.2B	1.2	0.852	original
Qwen-2.5-0.5B	0.5	0.901	original

Table 9: The seventeen models. Ordered by adapted news BPB. "original" marks the eleven models evaluated on all four corpora under the block readings too (Tables 5, 3, 6 and 10); "added" marks the six models added for the news corpus on size-adjacency and publisher-spread criteria fixed before they ran. All seventeen enter the seed replicates; all except Gemma-4-31B and Qwen-3.5-35B-MoE enter the fine-tuning experiment. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters; parameter count here is always the total.

	News		Reddit		Hacker News	
Model	unadapted	adapted	unadapted	adapted	unadapted	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LFM2.5-1.2B	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 10: BPB of the eleven models on the general-text corpora. Lower is better. Beginning-of-document tokens are context only and never scored. Rows are ordered by adapted news BPB. The leaderboard is maintained at https://github.com/DavidAfonsoValente/entropy-bench/blob/main/LEADERBOARD.md.

Three checkpoints are excluded. The two OLMo-2 models fell under a rule fixed in advance for infrastructure faults: their BPB cells exist, but the generation engine shared by the benchmark and fine-tuning evaluations would not initialise for them, so they have no benchmark or fine-tuned scores. The third, Qwen-2.5-32B, did not finish its 250 steps inside the run window.

F The Benchmark Re-Score

Harness configuration. MMLU-Pro is 5-shot chain-of-thought with exact match on one stratified 1,000-question subset (seed 42); HellaSwag is 10-shot length-normalised multiple-choice accuracy on all 10,042 validation examples; GSM8K is 5-shot chain-of-thought with flexible numeric extraction on all 1,319 test

examples. All bf16, greedy, with each tokenizer's own beginning-of-sequence token. The raw harness logs for all three benchmarks are released.[6]

One host. Generative scores do not pool across hosts: identical released weights move −0.60 to +0.69 GSM8K points when only the GPU changes from A100 to H100. The benchmark columns were therefore scored again in full on one host (A100-SXM-64GB) with one software build: seventeen models × three benchmarks = fifty-one cells, with the design and the exclusion rule fixed in advance. The task definitions match the earlier scoring exactly (identical task hashes for all seventeen models on GSM8K and HellaSwag; MMLU-Pro ships none), and every overlapping cell agrees with it to within 1.3 points, most to within 0.3.

Decoding settings. The newer harness honours each checkpoint's shipped `generation_config.json`, and the earlier one did not. Replacing every checkpoint's generation settings with one identical policy also discarded `suppress_tokens`, which is set for Gemma-4-12B; that omission cost it 9.86 GSM8K and 14.30 MMLU-Pro points, while HellaSwag, scored by likelihood, was unchanged. Keeping each checkpoint's own settings for all seventeen models reproduces the earlier scoring for Gemma-4-12B to within +1.21 points on GSM8K, +0.10 on MMLU-Pro and −0.27 on HellaSwag. The GSM8K cell is the one outside the 1.0-point tolerance fixed in advance, within the allowance of one such cell.

[6] https://github.com/DavidAfonsoValente/entropy-bench/tree/main/results